Investigation of antibacterial interaction of black phosphorus nanosheets

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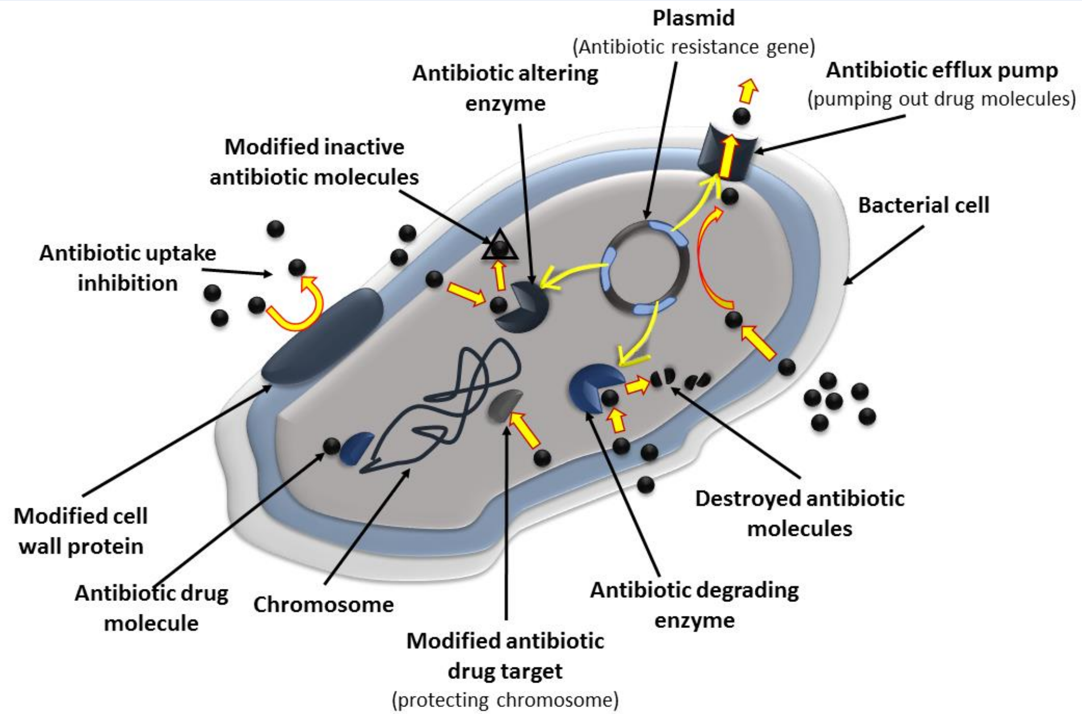
Contents

Need of the project – emerging antimicrobial resistance

- The state of the art evolving antimicrobial resistance (AMR) levels has become a global challenge.
- Osseointegration between the implant and tissues will deteriorate due to the presence of bacteria in addition to the spread of bacteria through-out the body.
- To counteract the mutating ability of bacteria, the research has been performed on studying the various nanomaterials interactions with bacteria based upon their shape, size, material, concentration, dispersion and interaction with bacteria.
- 2D black phosphorous (BP) is an emerging new material with implications for regenerative engineering. Bacterial infection during implant surgeries has been a major concern for the long-term stability of the installed implant due to the health and cost burdens on the patients, which can be addressed efficiently with black phosphorus nanosheets.

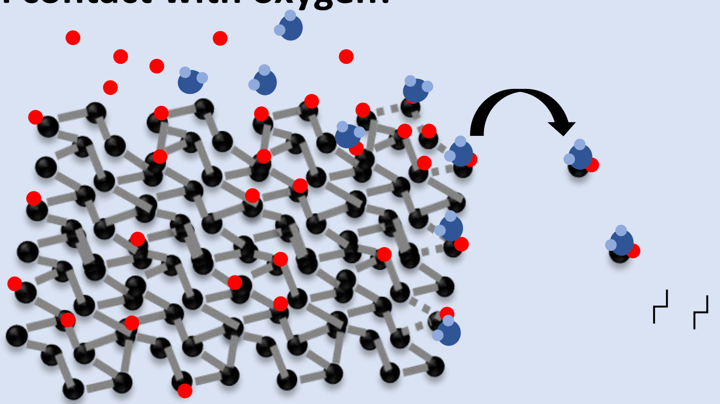


Note: Image is reserved for the publication.

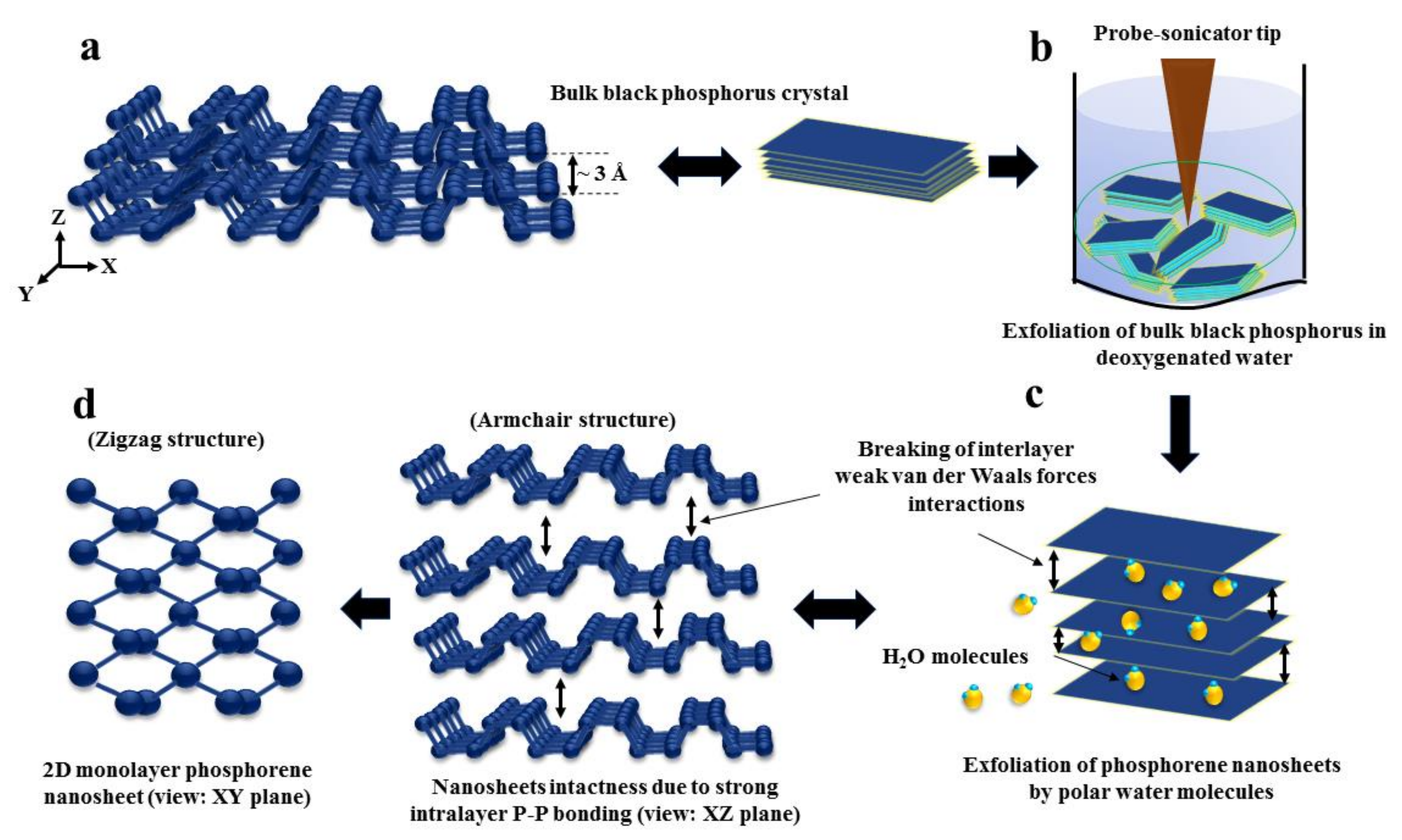
Black phosphorus crystals

https://www.ebay.com/itm/1-gram-high-purity-layered-black-Phosphorus-Phosphorene-crystals-99-998/122635257738

Challenge – oxidation of nanosheets in contact with oxygen!

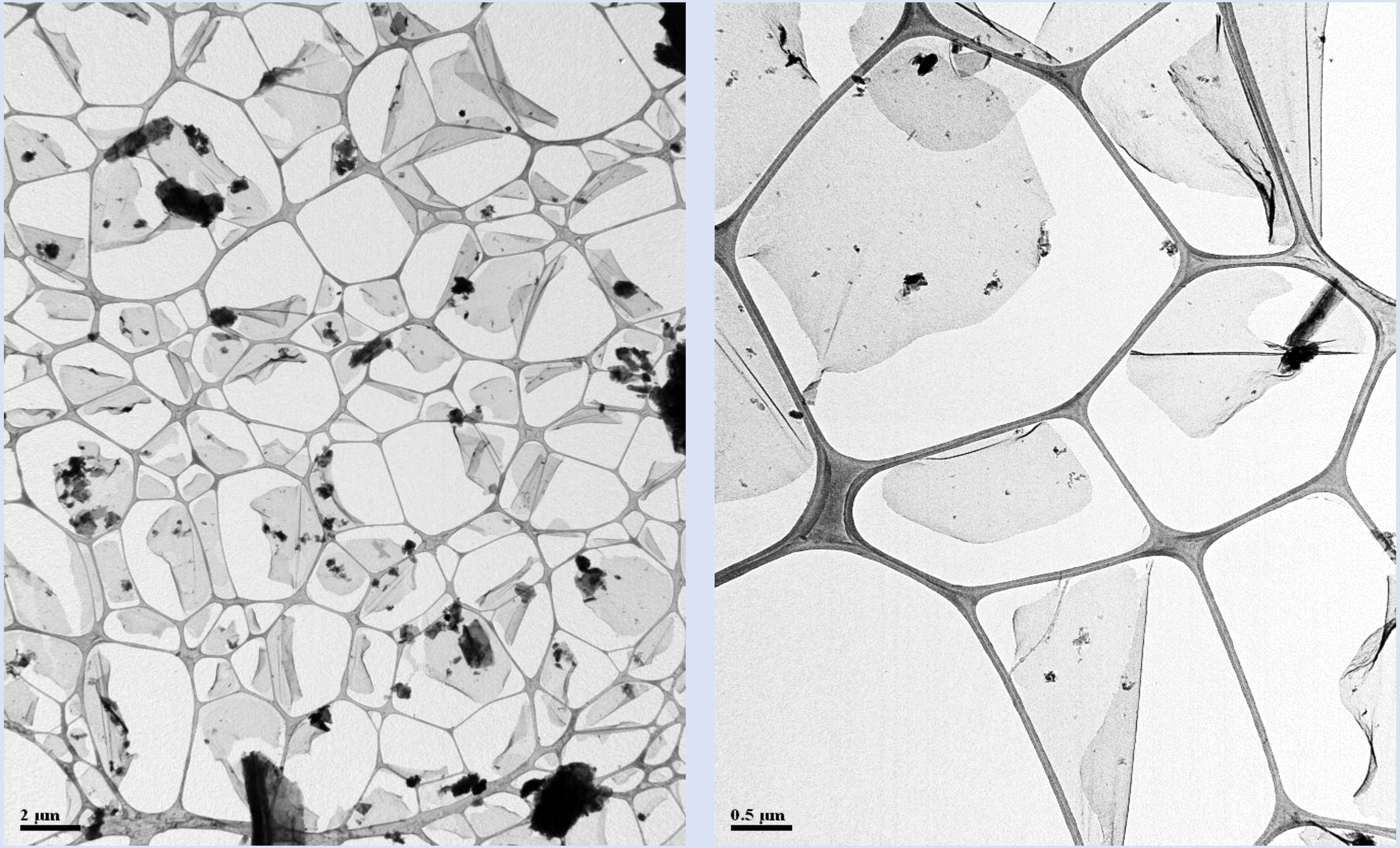


Synthesis of black phosphorus nanosheets by liquid exfoliation method



Note: Data is reserved for the publication.

Low-mag TEM images of black phosphorus nanosheets



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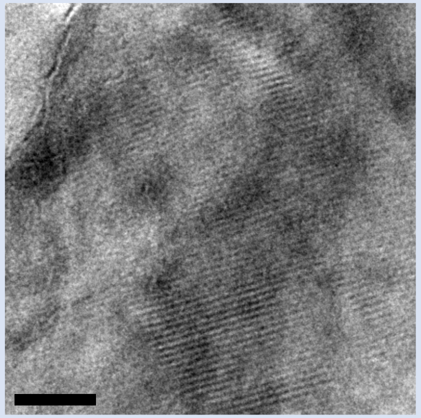
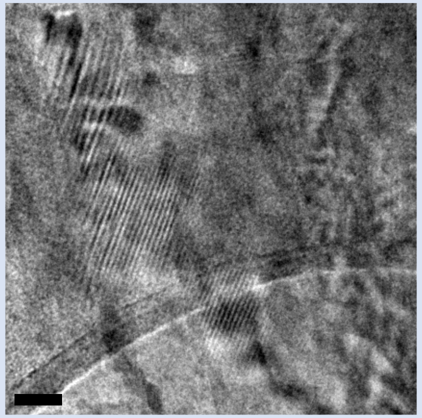
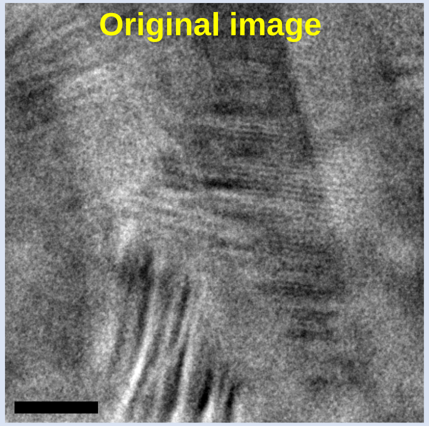
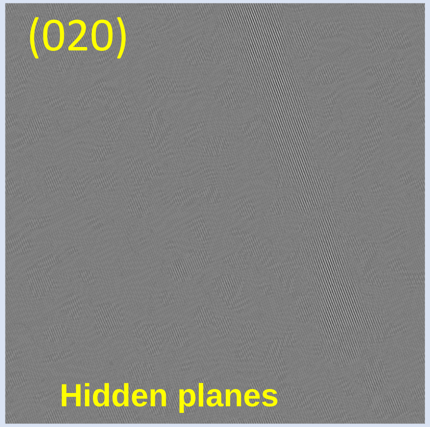
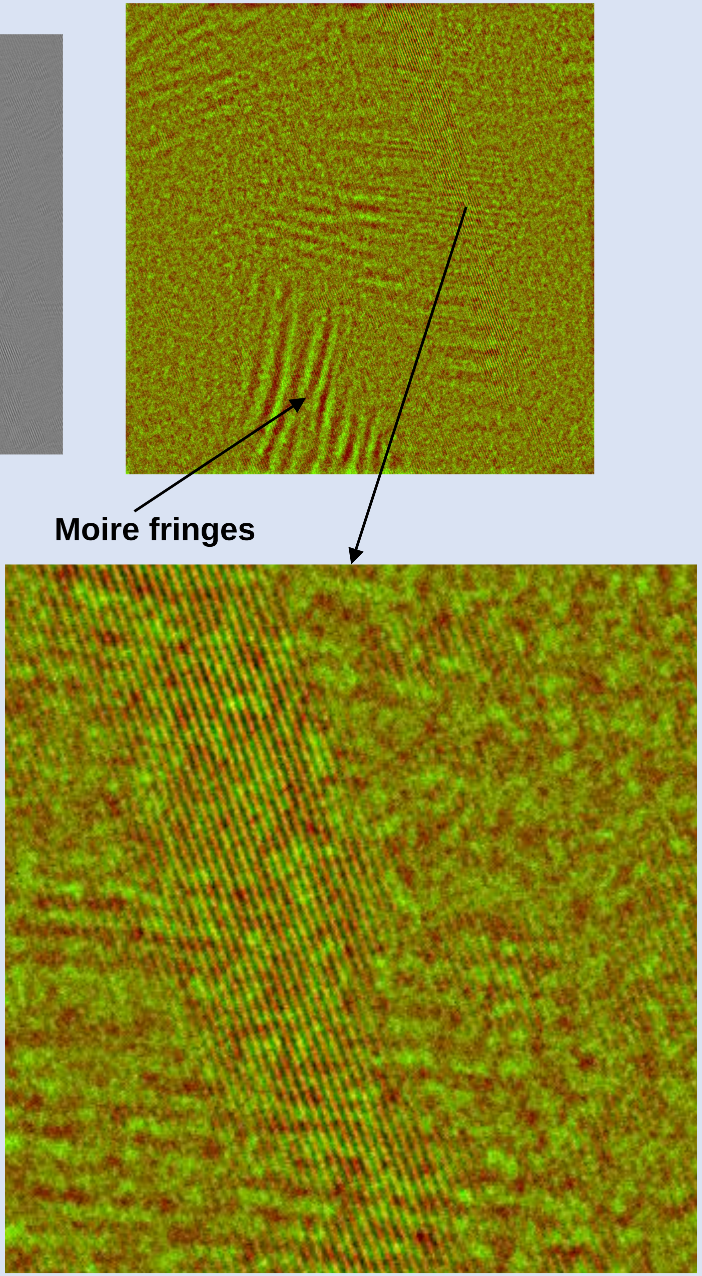
Challenges involved in HR-TEM images of black phosphorus nanosheets



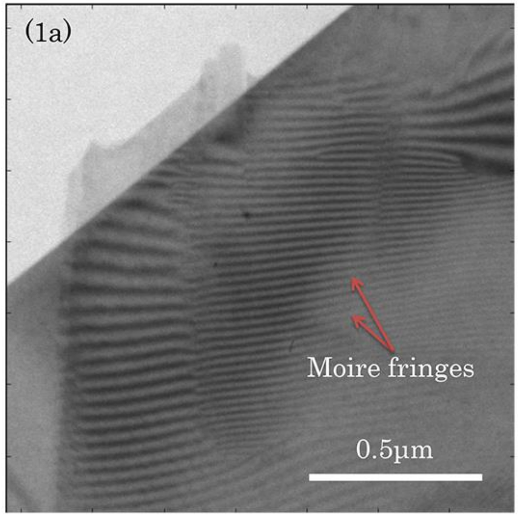


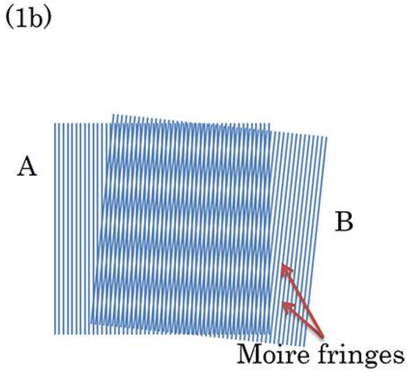






Moire fringes concept



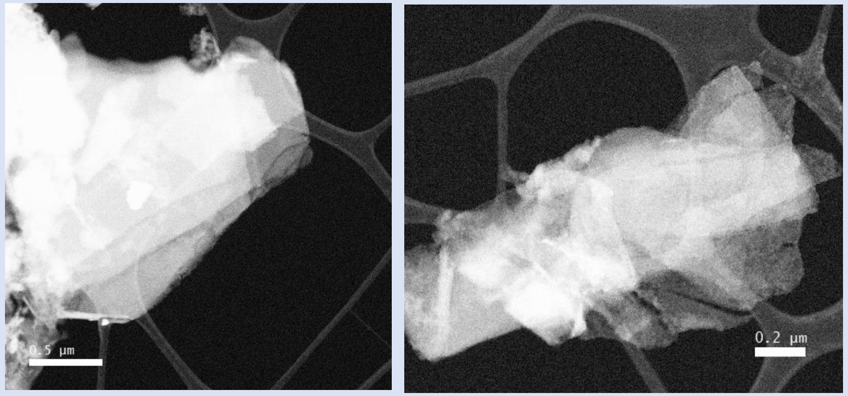


(1a) Bright-field image of mica showing Moiré fringes, taken at an accelerating voltage of 200 kV.
(1b) Schematic of rotation Moiré fringes. The fringes with an enlarged spacing appear when crystal lattices A and B which have the same spacing are superposed with a small orientation change.

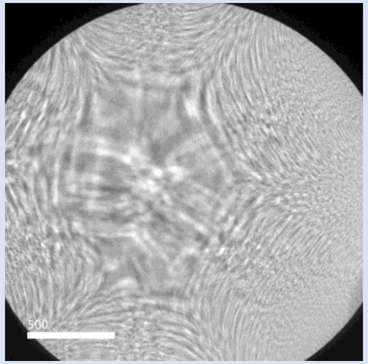
https://www.jeol.co.jp/en/words/emterms/search_result.html?keyword=Moire%20fringe

Note: Data is reserved for the publication.

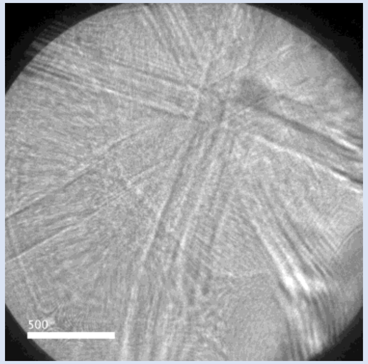
Challenges in annular dark field STEM images of black phosphorus nanosheets



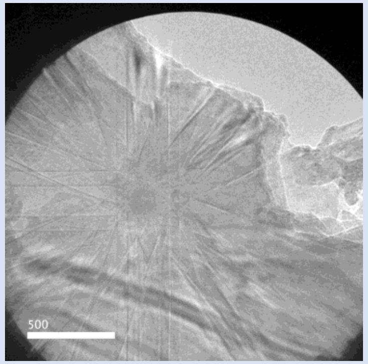
HAADF STEM image of phosphorene nanosheets



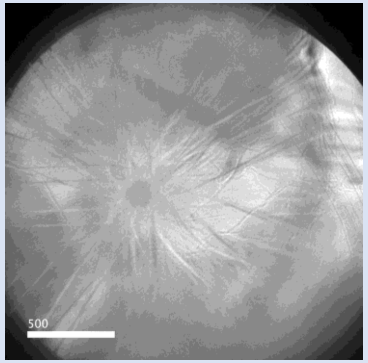
Ronchigram



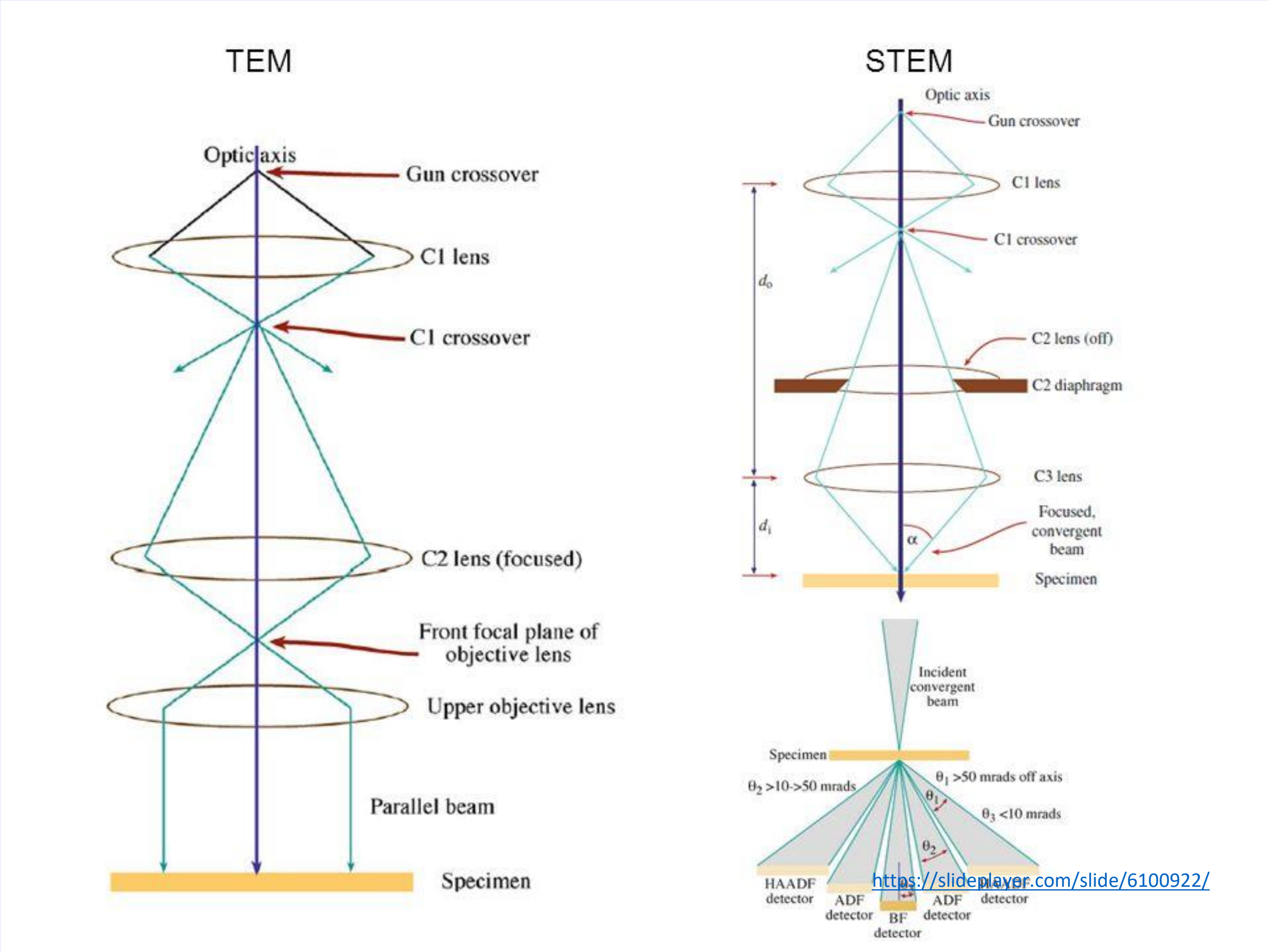
Kikuchi pattern



Proper tilting -zone axis



Rapid contamination in STEM mode

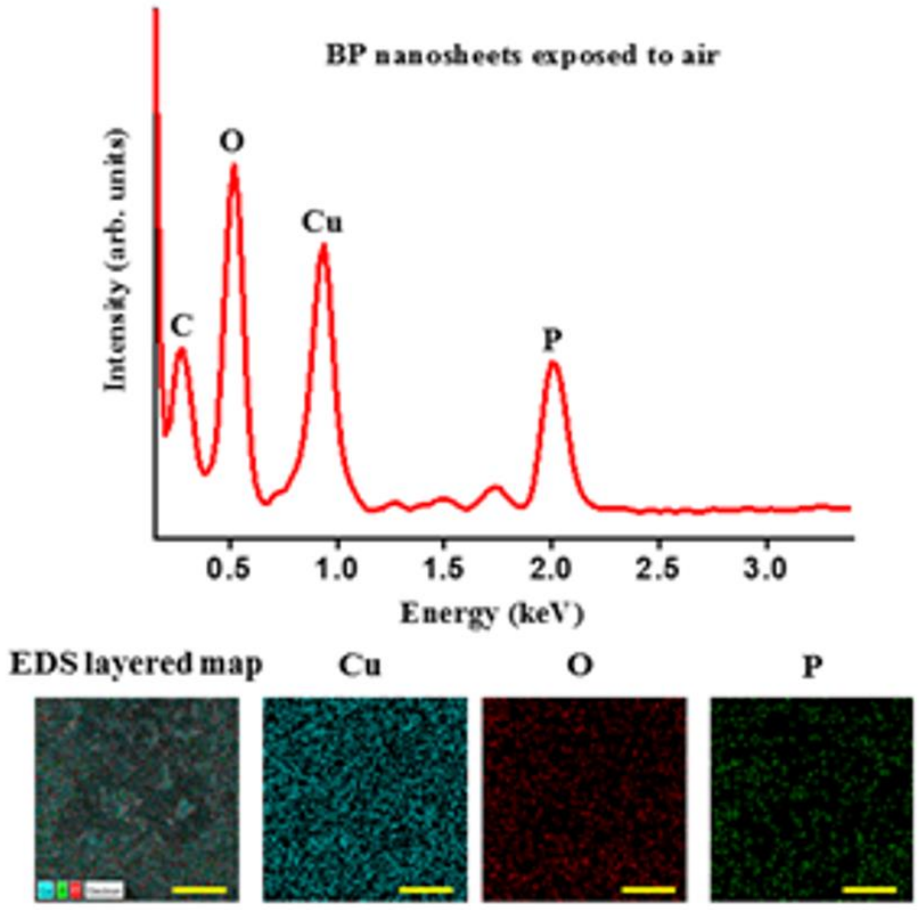


https://slideplayer.com/slide/6100922/

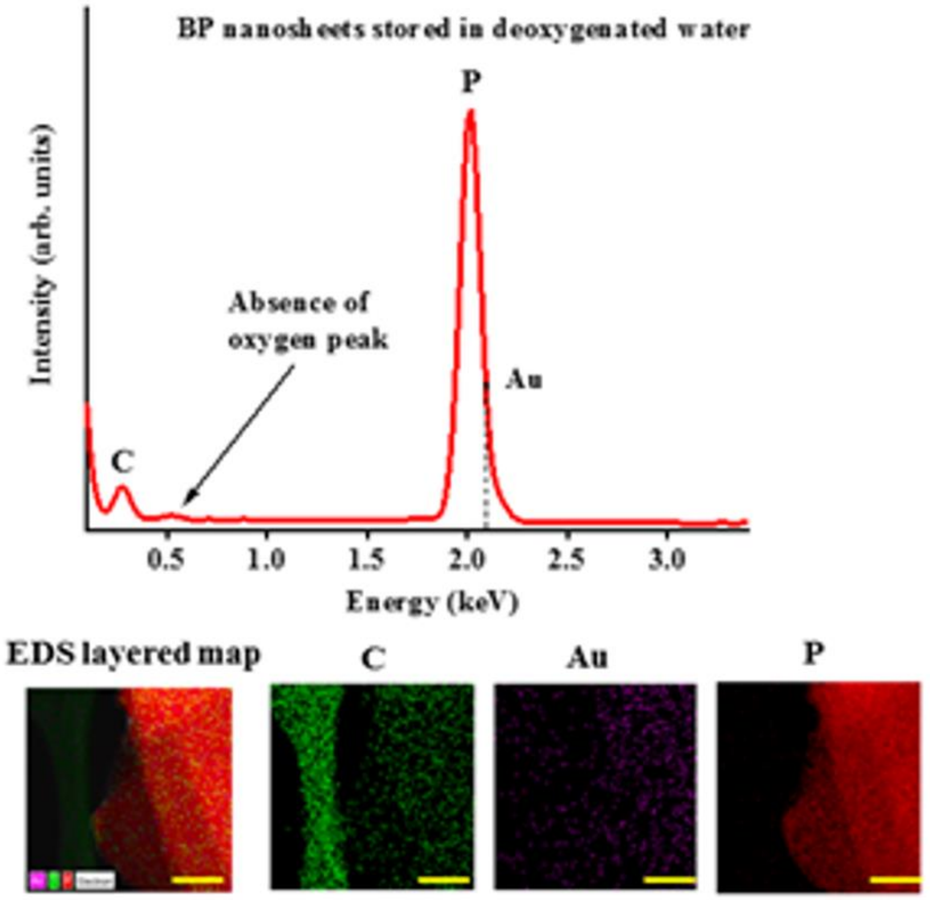
Note: Data is reserved for the publication.

Energy dispersive X-ray spectroscopy (XEDS) analysis of black phosphorus nanosheets

Challenge – oxidation of nanosheets in contact with atmospheric oxygen!



As-synthesized sheets were pristine before antibacterial studies!



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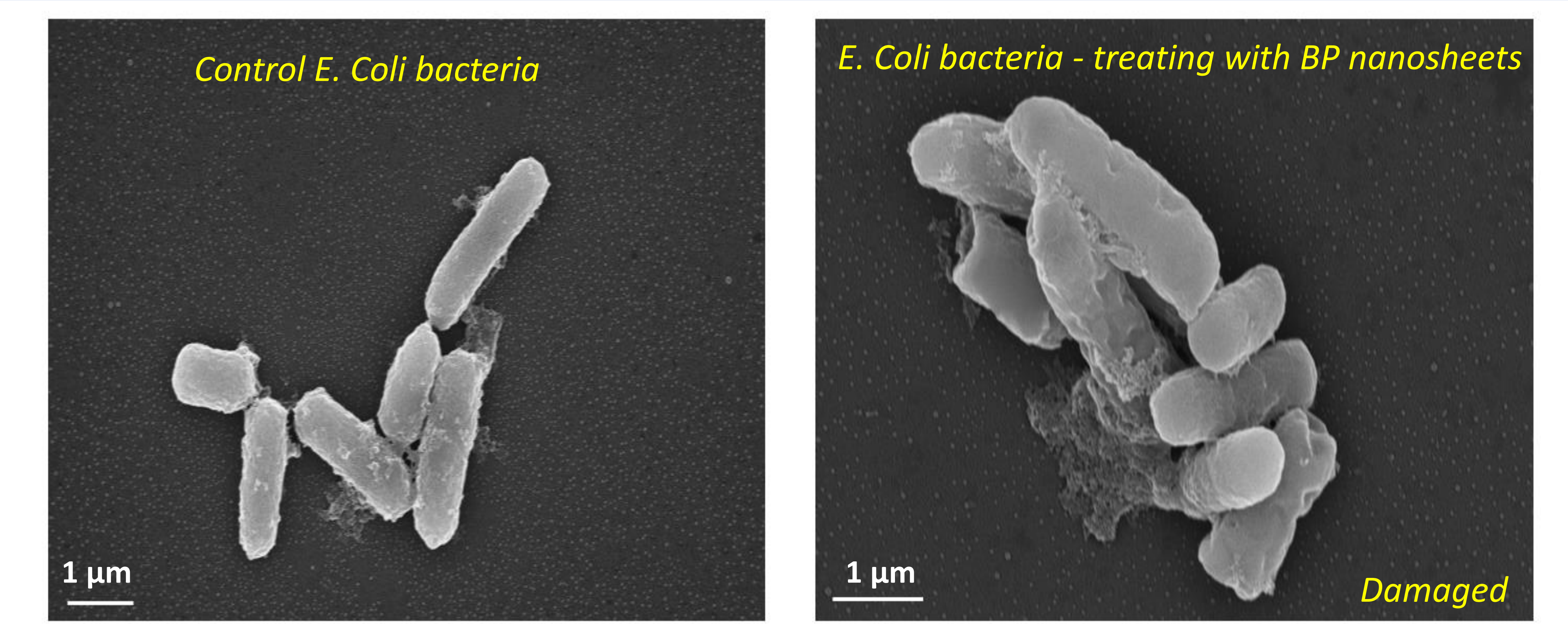
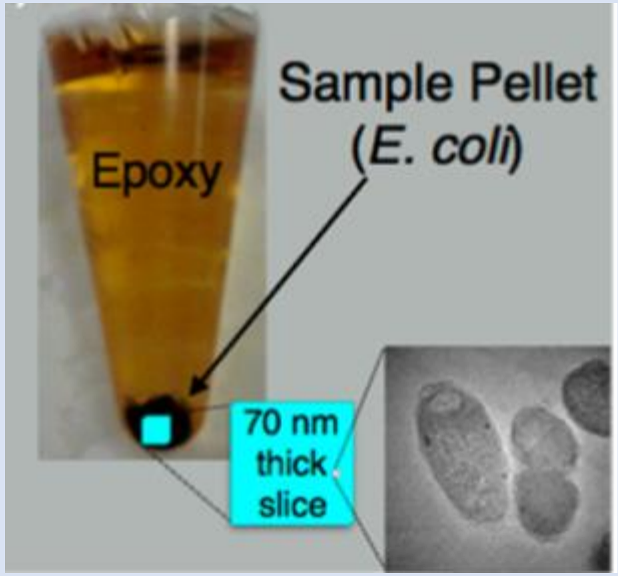
SEM analysis of ruptured bacterial cell upon treating with black phosphorus nanosheets



SEM observations of the morphological changes in *E. Coli* bacteria after treating with exfoliated BP nanosheets for 3 hours

Note: Data is reserved for the publication.

E. Coli bacteria ultramicrotome sample preparation for transmission electron microscopy studies



Ultramicrotome device

Preparing Embedded TEM Specimens for Sectioning

https://www.youtube.com/watch?v=yq77AB_8B4w

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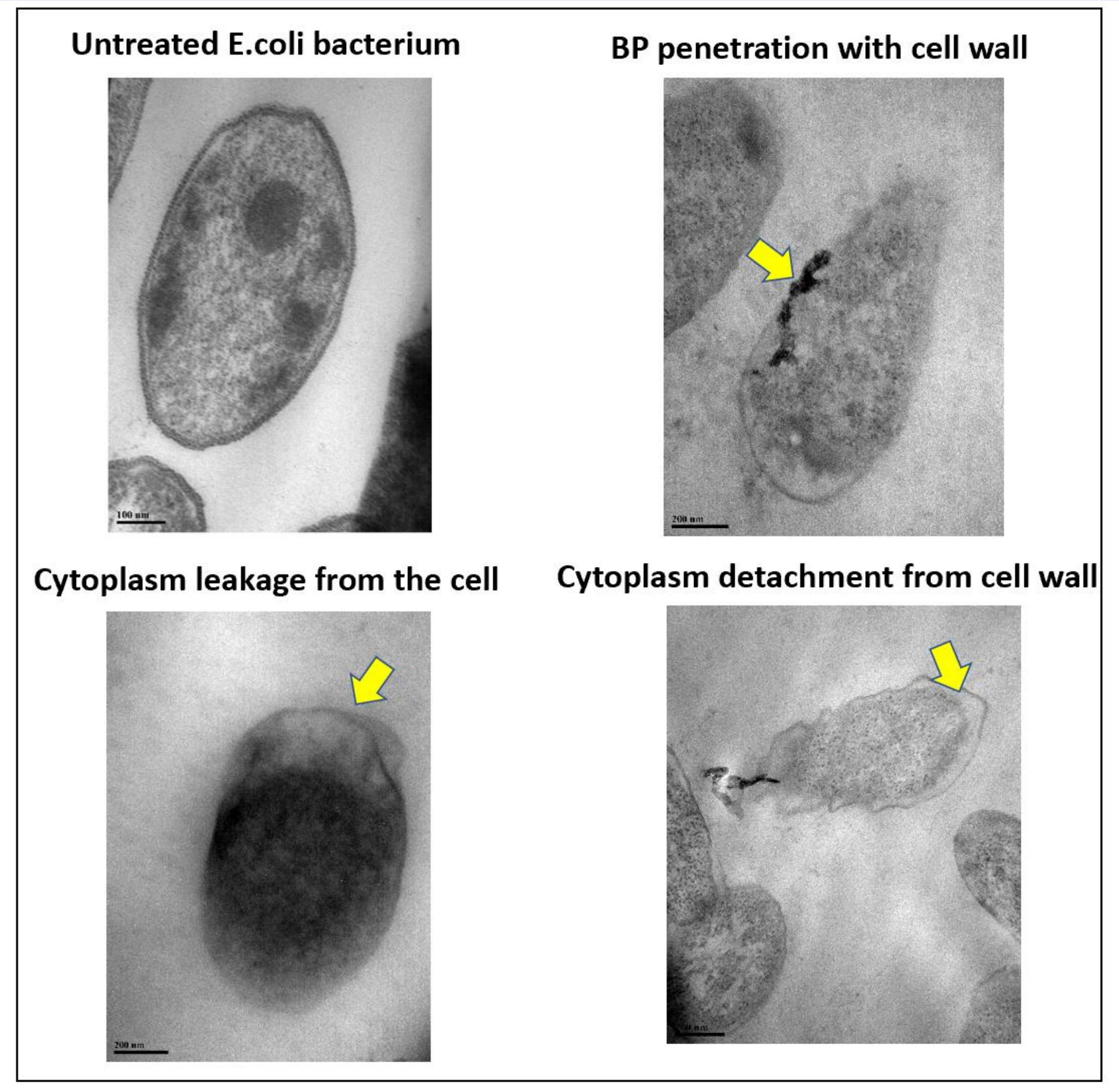
TEM analysis of ruptured bacterial cell upon treating with black phosphorus nanosheets



Confirmation of bacterial damage pathways!

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Conclusions

- Deoxygenated water is a reliable solvent for the exfoliation of ultrathin and the large surface area phosphorene nanosheets.
- Phosphorene (2D-BP) can maintain chemical stability without oxidation in the deoxygenated water.
- Black phosphorus nanosheets can kill bacteria synergistically upon causing physical damage.

Thank you!